An explicit bounded-R bridge for the complete C1 certificate

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The source formulas, full parent, and root identities below are those of `latex/certificates/c1_short.tex`, (P1), (BE3)–(BE9), (SP3), (SP6), and the twelve- and six-root exports in `extracted-lifts/local-order669/`, at commit `fb58c232e374dadf3cbb5f23e7e7413d429de884`. The new result is a quantitative extension of their source admission from $z \le 1$ to every specified finite R. No coefficient map or previously evaluated pencil is regenerated.

Theorem 1 (Explicit bounded-R admission)**.** *For a real $R \ge 1$, put*

$$B = 1 + R, \qquad \varepsilon_R = 2^{-40} B^{-8}, \qquad \delta_R = 2^{-20} B^{-2}.$$

Take exactly the native source formulas (P1), with

$$0 < x, \sigma \le \varepsilon_R, \quad 0 < z \le R, \quad 0 < b, h < 1, \quad r = bz < 1, \quad E, M, J > 0, \quad D, \beta \ge 0.$$

The complete parent F_w has a separated source/testing sum of squares on the independent rectangle

$$\sqrt{|u|}, \ |w| \le \delta_R$$

for every real state (W, X, Y, Z) and unrestricted $H = h/\sigma$. The same algebraic root identities and feature supports apply, including both state normals, both circuit-sign sectors, $D = 0$, and $\beta = 0$.

One fixed family for each R has at most 220 functions on $\lambda > 0$ and 157 on $\lambda \le 0$. Their source coefficients are given by the matrix splits below and the already evaluated retained-root formulas. Every numerical bound here is a real inequality, so the assertion holds over real closed source and testing fields as well.

The theorem does not assert that the functions are independent of R. It covers positive opening variables; a literal zero source opening requires its own chart. These are the same restrictions as the native bounded-positive-z C1 chart.

1 Source estimates that retain the exact identities

Write $Q = 1-b$, $\theta = 1-h$, $e = xh/\sigma$, $E = 1-e$, $K = 1+x$, $R_s = 1+xr$, $M = R_s - xz\theta$, and $J = 1+xh-\sigma(1-r)$. The positive denominator is

$$P = \sigma + K + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h \theta z.$$

The three opening-cleared coefficients are

$$p = \frac{z(1+xh)E + (1-r)(1+xhz)}{2P}, \quad l = \frac{(1+\sigma r)\{eR_s(1+\sigma z) + ME\}}{2P}, \quad \kappa = \frac{K(1+\sigma r)(1+\sigma z)}{2MP}.$$

As in (P1), $m = \kappa x z e$, $a = p/b$, $\ell = l/Q$, and $C(1-r) = Kz(1+\sigma r)E/(2P)$; D and β are the native nonnegative rational coefficients, not independently chosen variables. Put

$$H_0 = 1 + zQ, \qquad B_s = P + xz(J - \sigma),$$

$$\mathcal{H} = R_s H_0 + z\{(1-r)xh(2+xr) - e[2 + x(1+h+r\theta)] + Ke^2\}.$$

Throughout the proof ε can be any number with $0 < \varepsilon \le \varepsilon_R$ that bounds both x and σ. In particular $B^j\varepsilon$ is small for every exponent $j \le 8$ used below.

The exact source identities remain

$$2\theta P\beta = hB_s - \sigma H_0, \tag{BE3}$$

$$2P(JD + x\theta\beta) = \sigma^2\mathcal{H}, \tag{BE4}$$

$$eB_s = xH_0 + \sigma\mathcal{H} - 2PJD/\sigma \le xH_0 + \sigma\mathcal{H}. \tag{1}$$

No rescaling of the source or testing arguments has changed these equations. The following conservative bounds suffice:

$$1 \le P \le 1 + 4B\varepsilon, \quad |J-1| \le \varepsilon, \quad |M-1| \le B\varepsilon, \quad 1 \le B_s \le 1 + 6B\varepsilon, \quad 1 \le H_0 \le B, \tag{2}$$

$$e \le 5B\varepsilon, \quad |\mathcal{H} - H_0| \le 32B^2\varepsilon, \quad 0 \le D \le 2B\sigma^2, \quad 0 \le \beta\theta \le h, \tag{3}$$

$$|p - H_0/2| \le 8B^2\varepsilon, \quad |l - 1/2| \le 4B\varepsilon, \quad |\kappa - 1/2| \le 6B\varepsilon, \tag{4}$$

$$\tfrac{1}{3} \le p \le B, \quad \tfrac{1}{3} \le l, \kappa \le 1, \quad m \le 5B^2\varepsilon^2, \quad C(1-r) \le z \le B. \tag{5}$$

To check these bounds, use $r < 1$ before estimating the positive summands of P. This gives (2). Initially $E > 0$ gives $e < 1$, and dropping the negative term in $\mathcal{H}$ gives $\mathcal{H} \le 4B$. Equation (1) then gives $e \le Bx + 4B\sigma \le 5B\varepsilon$. Direct expansion yields

$$|\mathcal{H} - H_0| \le 4B\varepsilon + 3Be + 2Be^2 \le 32B^2\varepsilon.$$

Now $\mathcal{H} \le 2B$, and (BE4) gives the bound on D; (BE3) gives $\beta\theta \le h$. For the bounds on p, l, the useful exact numerator differences are

$$2Pp - H_0 = xzh(2 - r - e) - ze,$$

$$\frac{2Pl}{1+\sigma r} = M + e\{xz\theta + \sigma z R_s\}.$$

They prove (4)–(5) by (2); the formula for κ is handled in the same way.

Retain the source definitions (SP3):

$$\nu = (2D - 2\beta\theta + h)/\sigma, \quad \lambda = (8Dh + 4\beta\theta^2 + 2h^2)/\sigma - 4bp\sigma z, \quad \lambda_0 = \lambda - 2H,$$

$$K_A = 4\kappa H - \lambda, \quad B_x = 8\kappa e, \quad R_0 = -4\{2(C+D)(r-1) + 2p + r - 1\}.$$

Then

$$|\nu| \le 2B, \quad |\lambda_0| \le 3B, \quad |\lambda_0 + 2\theta H_0| \le 64B^2\varepsilon, \tag{6}$$

$$|K_A - (2\theta H_0 - 2h(1-r))| \le 1024B^4\varepsilon, \tag{7}$$

$$|R_0| \le 32B(x+\sigma), \quad |B_x| \le 40B\varepsilon, \quad |R_x| \le 16, \quad |A_y| \le 32B, \quad |\chi| \le 64B, \quad 0 \le k_1 \le 16. \tag{8}$$

Here R_x, A_y, χ, k_1 have precisely their printed (SP3) definitions. For (6), use the exact rearrangements

$$\nu = 2D/\sigma + [H_0 - ez(J - \sigma)]/P,$$

$$\lambda_0 = 2e\theta z(J-\sigma)/P - 2\theta H_0/P + 8Dh/\sigma - 4pr\sigma.$$

The only apparently unbounded factor in (7) is removed by

$$K(1+\sigma r)(1+\sigma z) - MP = xF_x + \sigma F_\sigma, \qquad F_\sigma = r - 1 + \sigma r z,$$

$$H(2\kappa - 1) = \frac{eF_x + hF_\sigma}{MP}.$$

The polynomial F_x, after $r = bz$, has coefficient ℓ^1 norm 37 and z-degree 3. Thus $|F_x| \le 37B^3$. Also $(MP)^{-1} \le 2$ and $|(MP)^{-1} - 1| \le 12B\varepsilon$, giving

$$|H(2\kappa - 1) + h(1-r)| \le 384B^4\varepsilon.$$

Together with $K_A = 2H(2\kappa - 1) - \lambda_0$, this proves (7). The pole-free identity

$$\begin{aligned} -PR_0/4 = {} & zE\{1 + xh - K(1+\sigma r)\} \\ & + (1-r)(1 + xhz - P) - 2DP(1-r) \end{aligned}$$

gives the first bound in (8); the others follow by substitution into (SP3), using $|h - Q| \le 1$ and $\beta\theta^2 \le h\theta \le 1/4$.

The downstream source identities. In particular $1/2 \le \mathcal{H} \le 2B$. Therefore the unchanged sphere definitions

$$d_0^2 = \frac{2PJD}{\sigma^2\mathcal{H}}, \qquad b_0^2 = \frac{2Px\theta\beta}{\sigma^2\mathcal{H}}$$

give $d_0, b_0 \ge 0$, $d_0^2 + b_0^2 = 1$ without dividing by either coordinate, and

$$\frac{\sigma^2}{8} \le JD + x\theta\beta \le B\sigma^2.$$

The exact (BE6)–(BE8) formulas for D, β, H, m_0, m_1 are unchanged. For example

$$H = \frac{H_0}{B_s} + \frac{\sigma}{x}\frac{\mathcal{H}}{B_s}b_0^2, \quad \frac{1}{6} \le \frac{m_0}{x^2 z} \le B, \quad \frac{1}{12} \le \frac{m_1}{x\sigma z} \le 2B.$$

The three (BE9) errors are bounded, respectively, by $8B\varepsilon, 8B^2\varepsilon, 4B\varepsilon$. Finally $2P(p-l)/z$ is a polynomial in (x, σ, z, b, h, e) with coefficient norm 18 and z-degree 2. Hence $|(p-l)/z| \le 9B^2$. This proves all the boundedness and asymptotic assertions used downstream, including the radial boundary estimate, from the same exact source identities.

2 A uniform bound for the positive-angular Gram

Suppose $\lambda > 0$. Use the existing twelve-root column

$$(N_x, N_y, R, \Gamma_*, A_\theta, T_\theta, d_1, d_2, d_3, d_4, U_*, K_*)^T$$

and its exact matrix $Q(s, w)$, where $s = \sqrt{u}$. Put

$$d_G = 4D/\sigma^2, \quad \rho = \lambda x/\sigma, \quad S = d_G + \rho, \quad L_0 = \lambda\sigma, \quad q_0 = \lambda\sqrt{x}, \quad c = \max(0, 1 - K_A), \quad a_0 = \sqrt{x}/\sigma.$$

The source identities and (2)–(8) imply

$$|S - 2H_0| \le 128B^2\varepsilon, \quad 1 \le S \le 3B, \quad 0 \le d_G, \rho \le 3B, \quad 0 < L_0 \le 3, \tag{9}$$

$$\frac{\sqrt{d_G}}{S}, \frac{\sqrt{\rho}}{S}, \frac{d_G}{S}, \frac{\rho}{S} \le 1, \qquad q_0/S \le 2, \quad \sqrt{q_0} \le 2B. \tag{10}$$

Indeed the exact identity to use is

$$S = 2\mathcal{H}/P + d_G(1 - J + 2hx) + 2eH_0/P - 2e^2z(J - \sigma)/P - 4prx.$$

Initially $d_G \le 8B$; bounding the five terms gives (9). Then $q_0^2 = L_0\rho$ gives (10). Further,

$$(x + \sigma)^2 H = xe + 2\sigma e + \sigma h \le 2\varepsilon, \qquad |R_0\sqrt{\lambda}| \le 128B\sqrt{\varepsilon}.$$

The second conclusion follows from $\lambda \le 2H + 3B$ and $(x + \sigma)^2\lambda \le 5\varepsilon$.

The compensation terms satisfy

$$0 \le c \le 4, \quad ca_0 \le 32B, \quad c/L_0 \le 32, \quad c/\lambda \le 32\sigma. \tag{11}$$

For $c > 0$, (7) forces $h > 1/8$, because the reference $2\theta H_0 - 2h(1 - r)$ is at least $3/2$ when $h \le 1/8$. Now $L_0 \ge h > 1/8$ and

$$a_0^2(hB_s - \sigma H_0) \le \mathcal{H} \le 2B, \qquad hB_s - \sigma H_0 \ge 1/16.$$

Thus $a_0 \le 8B$. If $c = 0$, every compensation term is zero; no unused denominator is bounded away from zero.

Here is a finite positive-coefficient check of every matrix entry. Let $\tau = \sqrt{\varepsilon}$, and regard the twelve root letters as independent nonnegative formal variables, abbreviating $G = \Gamma_*$, $A = A_\theta$, $T = T_\theta$, $U = U_*$, $K' = K_*$. The two full normal conductors $\mathfrak{U} = u\Gamma$ and $\mathfrak{V} = (R_0 - 8pu^2)qw\Gamma$ have coefficient majorants

$$f_1 = uG + \tau d_1 + ud_2 + (u + \tau)d_3 + \tau T,$$

$$f_2 = 64B^2\tau^2 wG + 64B^2\tau^4 R + 64B^2\tau^2(w+\tau)(d_2+d_3) + (128B^2\tau^2 + 128B^2\tau w)T + 8B^2 uw f_1.$$

These follow by substituting (10) in the two exact conductor formulas in `twelve_root.tex`; in particular no normal coupling is deleted. Let B_{src} denote the original source reference with N_x entry 2, radial form $J_C^2 + J_\ell^2$, G entry 1, angular entries $(K_A + c)/2$, four circuit entries $1/2$, and final entries $1, 1/2$. Then $B_{\mathrm{src}} \succeq I_{12}/3$. The correction $Q - B_{\mathrm{src}}$ is coefficientwise dominated by

$$\begin{aligned}\mathcal{P} = {} & (\tfrac{131}{2}B\tau^2 + 4u^2)R^2 + (64B^2\tau^2 + 16Bu^2 + 8w^2)N_y R \\ & + 8BN_x f_1 + (N_y + R)f_2 + 8BuwN_xN_y + 16BuwN_xR \\ & + 40B^2\tau^2 wN_x A + 32BuN_y A + 64BuAR + 2BsUR \\ & + 32BuU^2 + 64BsUK' + 32B\tau(K')^2 \\ & + [(64B+32)u + 32\tau]G^2 + (64\tau^2 + 4u + 4\tau)\sum_i d_i^2.\end{aligned}$$

For example, $|E^0_{RR}| \le 64B\varepsilon$, the coefficients of E_{yR} on $(1, u^2, w^2)$ are at most $(64B^2\varepsilon, 16B, 8)$, and the three compensation expressions are bounded using (11). These give exactly the displayed majorant.

The sum of its coefficients in all variables other than B is

$$1480B^2 + \tfrac{887}{2}B + 364 < 2048B^2 \qquad (B \ge 2).$$

Every term with testing degree zero contains a factor τ. For a quadratic form, total absolute coefficient mass bounds the matrix row norm: an off-diagonal coefficient is split equally between two matrix entries. Therefore, on the exact support

$$I = \{(0,1),(0,2),(1,0),(2,0),(2,1),(4,0),(4,1)\},$$

$$Q = Q_0 + \sum_I C_{ij} s^i w^j, \qquad \|Q_0 - B_{\mathrm{src}}\|_{\mathrm{row}} \le 2048B^2\sqrt{\varepsilon}, \qquad \|C_{ij}\|_{\mathrm{row}} \le 2048B^2. \tag{12}$$

The 220-function organization puts the four already retained positive terms

$$2z[(1+2m)u^2 + (1+2D)w^2]N_y^2 + [zk_1u^2 + 2zb^2(2D+1)w^2]R^2$$

into the main matrix. Their additional majorant is $4B(u^2+w^2)N_y^2 + (16Bu^2 + 4Bw^2)R^2$. The resulting mass is $1480B^2 + 943B/2 + 364 < 2048B^2$. Thus (12) proves the same guard for this organization as well. This is a regrouping of four existing squares, not a new guessed Gram space. The old $z \le 1$ row estimate for w^2 is 5, not 4: $2z(1+2D)(1+b) \le 4(1+6\varepsilon^2) < 5$.

Set $M_R = 2048B^2$, $\delta = \delta_R$, and

$$T_\delta = 2\delta + 2\delta^2 + \delta^3 + \delta^4 + \delta^5.$$

At the stated constants,

$$2048B^2\sqrt{\varepsilon_R} \le 1/2048, \qquad M_R T_\delta \le 7M_R\delta = 7/512 < 1/48.$$

Consequently the explicit source matrices

$$H_0^{\mathrm{mat}} = Q_0 - M_R T_\delta I \succeq I/6, \qquad H_{ij}^{\pm} = \frac{\delta^{i+j}}{2}(M_R I \pm C_{ij}) \succeq 0$$

give the exact split

$$Q = H_0^{\mathrm{mat}} + \sum_I \left[H_{ij}^{+}\left(1 + \frac{s^i w^j}{\delta^{i+j}}\right) + H_{ij}^{-}\left(1 - \frac{s^i w^j}{\delta^{i+j}}\right)\right].$$

Every scalar factor is nonnegative on $0 \le s, w \le \delta$. The circuit remainder has coefficient

$$q_C = \tfrac{1}{2} - (cd_G/S)(2/\lambda + \sqrt{x} + u).$$

Its source constant after subtracting $(cd_G/S)\delta^2$ is at least $1/2 - 64\varepsilon - 4\sqrt{\varepsilon} - 4\delta^2 > 1/3$. Thus the existing circuit split with $\sqrt{\delta^2 - u}$ is also valid. Source-field square completion proves the asserted separated SOS.

3 The nonpositive-angular branch needs no new shear

Suppose $\lambda \le 0$, and put $L = \lambda/\sigma$, $s_6 = \sqrt{1+L^2}$. Nonnegativity of the four terms defining $\lambda + 4pr\sigma$ gives

$$H^2 \le 2pr \le 2B, \quad h \le 2B\sigma, \quad \beta\theta^2 \le B\sigma^2, \quad -4B \le L \le 0, \quad s_6 \le 5B.$$

Hence $\theta \ge 1/2$, $\beta\theta \le 2B\sigma^2$. (BE3) gives $H \ge H_0/B_s \ge 3/4$; therefore $4\kappa H \ge 1$, while $4a, 4\ell \ge 1$ follows from (5). Finally (BE4) gives

$$d_G = \frac{2\mathcal{H}}{PJ} - \frac{4x\theta\beta}{J\sigma^2} \ge \frac{2(1-32B^2\varepsilon)}{1+6B\varepsilon} - \frac{8B\varepsilon}{1-\varepsilon} \ge 1.$$

After multiplying positive denominators, the last inequality is

$$1 - (64B^2 + 14B + 1)\varepsilon + (16B^2 + 6B)\varepsilon^2 \ge 0;$$

its left side is at least $1 - 72B^2\varepsilon > 0$.

Use the unchanged six-root identity (SR1)–(SR3), on $v = (N_x, N_y, R, \Gamma, A_\theta, s_6 uw(u^2 - x)W)^T$. Its source reference G_0 has N_x entry 2, radial block $\left(\begin{smallmatrix} 2 & 2b-1 \\ 2b-1 & b^2+Q^2 \end{smallmatrix}\right)$, angular entry $4\kappa H$, and $(\Gamma, \widehat{V})$ block $\left(\begin{smallmatrix} d_G & L/(2s_6) \\ L/(2s_6) & 1 \end{smallmatrix}\right)$. The displayed guards imply $G_0 \succeq I_6/3$.

The coefficient matrices printed in (SR2) satisfy, directly,

$$\|E_{00}\|_{\mathrm{row}} \le 128B^2\varepsilon, \qquad \|E_\gamma\|_{\mathrm{row}} \le 128B^2 \quad (\gamma \in \{10, 01, 11, 20, 02, 21, 30\}).$$

For an explicit row check, the seven nonconstant rows are bounded, respectively, by

$$49B, \quad 64B^2\varepsilon, \quad 12B, \quad \max(22B, 25B^2\varepsilon), \quad 8B, \quad 8B^2, \quad (25/2)B^2\varepsilon.$$

For E_{00} use the previously proved constant radial estimates and $s_6 x/2 \le (5/2)B\varepsilon$. Thus

$$\|E_{00}\|_{\mathrm{row}} < 1/12, \qquad \delta^{|\gamma|}\|E_\gamma\|_{\mathrm{row}} \le 128B^2\delta = 1/8192 < 1/56.$$

The same source matrices (SR6), with $\eta = 1/56$ and $y_\gamma = u^{\gamma_1} w^{\gamma_2}/\delta^{|\gamma|}$, apply on $0 \le u, w \le \delta$, and hence on the common smaller rectangle $0 \le \sqrt{u}, w \le \delta$. The residual source matrix is $G_0 + E_{00} - I/8 \succeq I/8$. This proves the complementary branch without assuming its guards. In particular $D > 0$ here; its literal zero face occurs in the positive branch. No inverse of β occurs in either construction.

4 Fixed functions, and what transfer does and does not give

The positive main family is the twelve pre-Gram functions

$$\mathcal{V} = \{X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^{3/2}W, u^{1/2}W, \sqrt{uwWZ_+}\}$$

times 1 and the fourteen factors $\sqrt{1 \pm s^i w^j/\delta^{i+j}}$, $(i,j) \in I$. First apply the fixed angle sign gauge, then the common state sign making $W \ge 0$; here $Z_+ = \max(Z, 0)$ and $Z_- = \max(-Z, 0)$. Add the seven ordinary functions

$$\{wX, uwZ, uX, u^2wW, uY, wY, uwY\},$$

the nineteen angular functions printed in `fixed-features/fixed_box_c1.tex`, and its seven circuit functions t_i together with $\sqrt{\delta^2 - u}\, t_i$. These are $180 + 7 + 19 + 14 = 220$ labelled functions; the nineteen angular and seven circuit formulas contain no R or new source choice. Their verified source-linear root prescriptions remain unchanged. The six-root family is its ten pre-Gram monomials

$$\{X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^3wW\}$$

times 1, $\sqrt{1 \pm y_\gamma}$, plus the same seven ordinary functions: $10 \cdot 15 + 7 = 157$.

Alternatively the previously used polynomial testing cover is

$$u = \delta^2 A_t^8, \quad w = \delta B_t^4, \quad \xi^2 = 1 - A_t^4, \quad \eta^2 = 1 - B_t^4,$$

with the same quarter-state cover and normalization equations. The existing geometric-sum identities use exactly the same sixteen multiplier monomials for every fixed R. Its primitive monomial product maps are unchanged.

The parent coefficient embedding contains the newly specified powers of δ; this statement does not identify it with the previously eliminated order-669 affine source pencil at its old numerical δ.

For rational or algebraic coefficients, one may replace a given real R by any integer $N \geq R$ and use ε_N, δ_N. Both constants are then rational, and every quarter-power used by the cover is algebraic.

All root identities are polynomial after the existing algebraic covers. Hence testing axes and zero state coordinates are literal specializations. The two exact angle gauges extend to both signs, and the two state sectors cover all states. The full (SP6), not a determinant restriction, is the identity assembled above.

For a bounded positive source z in a real closed extension, some real R bounds it. Infinitesimal x, σ are below this ε_R, and infinitesimal tests are below this δ_R. The finite inequalities therefore recover the corresponding bounded-z infinitesimal admission assertion in B.23 by transfer. The quantifiers are

$$\forall R \geq 1 \quad \exists(\varepsilon_R, \delta_R, \mathcal{V}_R) \quad \forall \text{ native sources and independent tests in the boxes.}$$

They do not supply one R-independent testing space. The displayed radical branch locations depend on δ_R; a finite cover of the remaining independent testing regions and source charts is still needed.

5 An exact obstruction to one proposed fixed-δ repair

The need for a new argument is visible in a two-by-two principal minor. This is an obstruction to the unchanged principal matrix, not a Bell-slack counterexample or a no-lift theorem.

Proposition 2. *For every fixed $u_* > 0$, arbitrarily large finite z admit native sources arbitrarily close to the source origin for which the unchanged twelve-root principal Gram at $(u, w) = (u_*, 0)$ is indefinite. Adding any amount of the positive J_C^2, J_ℓ^2 radial reserves, or applying an invertible source-only diagonal congruence, cannot repair this principal minor.*

Proof. It suffices to take $z \geq 1$ and

$$r = \tfrac{1}{2}, \quad b = \frac{1}{2z}, \quad H_0 = z + \tfrac{1}{2}, \quad \sigma = t, \quad x = t^2, \quad h = H_0 t, \quad 0 < t \leq \varepsilon_z.$$

Here $B = z + 1$. Every strict native gap is positive at this bound: $h < 1$, $E = 1 - H_0 t^2 > 0$, and $M, J > 0$ by (2), or directly from their formulas. Since $J - t > 0$ and $P > 1$, $B_s = P + t^2 z(J - t) > 1$, so (BE3) gives $\beta > 0$ exactly. The numerator $N_D = K(1 + tz)(1 + t/2) - JP$ of D satisfies

$$N_D = t^2(H_0 + \mathcal{R}), \qquad |\mathcal{R}| \leq 19B^4 t^2.$$

Indeed $\mathcal{R}$ is a polynomial whose monomials all have t-degree at least two, whose z-degree is at most four, and whose absolute coefficient sum is 19; the checker prints its derivation. Thus $D > 0$. Also

$$\lambda \geq 2t(H_0^2 - p) > 0,$$

because $p \leq B$ and $H_0^2 - B = z^2 - 3/4 > 0$. Finally $c = 0$ by (7), as its reference tends to $2H_0 \geq 3$ and its error is bounded explicitly by $1024B^4\varepsilon_z$.

The exact leading coefficients are

$$\beta \sim \tfrac{1}{2}H_0(z + 1)t^2, \quad D \sim \tfrac{1}{2}H_0 t^2, \quad \lambda \sim 4H_0^2 t,$$

$$\nu \longrightarrow H_0, \quad d_G, S \longrightarrow 2H_0, \quad \rho \longrightarrow 0.$$

At $w = 0$, the (N_x, Γ_*) principal submatrix is exactly

$$\begin{pmatrix} 2 & -2\nu\sqrt{d_G}\, u_*/S \\ -2\nu\sqrt{d_G}\, u_*/S & 1 \end{pmatrix},$$

whose determinant tends to $2 - 2H_0 u_*^2$. For $H_0 u_*^2 > 1$ it is negative for small enough t. An entirely explicit choice for $0 < u_* \leq 1$ is $z = 32/u_*^2$, $0 < t \leq \varepsilon_z$. The exact formula for ν gives $|\nu - H_0| \leq 16B^2 t$; also $\rho = \lambda t \leq 5Bt$ and $|S - 2H_0| \leq 128B^2 t$. Consequently $\nu \geq H_0/2$, $d_G \geq H_0$, $S \leq 3H_0$, and thus

$$\det \leq 2 - H_0 u_*^2/9 < 0.$$

The radial reserves affect only the (N_y, R) rows, and an invertible congruence preserves inertia. This proves the claim. □

This obstruction must not be overstated. The separately retained averaged square $(L_0/2)F_A^2$ can change the defective minor:

$$F_A = u(Z - a_0 wW) = \frac{u}{\sigma}\Gamma + \frac{uw}{\sigma kq}T_\theta.$$

Along the path above at $w = 0$, its contribution to the Γ_*^2 coefficient tends to $H_0 u^2$, precisely the scale missing from the unchanged principal matrix. A polynomial completion that reorganizes this square therefore remains open. What fails is the proposed repair by radial reserves and constant diagonal normalization alone.

Reproducibility. `check_bridge.py` uses exact SymPy arithmetic to check the new positive majorants, their support and coefficient masses, the two denominator removals and radial divisibility, the reserve inequalities, the nonpositive d_G guard, and the explicit path's leading coefficients. Its output is `check_results.json`. It also evaluates the explicit rational source with $u = 2^{-40}$, $z = 2^{85}$, and $t = 2^{-40}(1 + z)^{-8}$. The native guards and the negative determinant are checked with rational arithmetic; the source and a rationally congruent 2×2 matrix are printed in `fixed_delta_obstruction.json`. It does not rerun the 682 old coefficient equations or regenerate any old affine pencil. The analytic admissibility proof uses the native identities throughout, not independent intervals for tied derived source quantities.